Project Description:

I started with the idea of a Bank System and, as we briefly discussed early on, shifted to a Digital Wallet. I remember thinking it would be challenging to implement, and it was, not because it required anything beyond what we learned in class, but because as I worked on it I kept wanting to improve it further, so it ended up being more extensive than I originally imagined. I genuinely enjoyed working on it and plan to keep improving it over time.

The idea was to build an application similar to Venmo, where a user has a wallet balance, can link payment methods to add funds, and can send and receive money with other people, as well as withdraw funds back to a linked bank account. The application features a graphical interface built with JOptionPane, allowing users to interact through dialog windows and buttons. Users can create an account with their name, email, phone, and password, and log back in at any time to find their wallet balance, transaction history, and linked payment methods exactly as they left them.

The requirements applied in the project were the creation of the BankAccount and CreditCard classes, which are forms of payment and extend the abstract class PaymentMethod. I also created a custom exception, InsufficientFundsException, to check whether there are sufficient funds for any type of transfer. In addition, I used a recursive method, readAmount(), to verify that the amount entered for a transaction is valid. Finally, the FileManager class was created to manage the .txt files as a simplified database, writing and reading user accounts, transactions, and payment methods so that sessions are not lost between uses.

Overall, the idea was a simplified simulation of how a real digital wallet works.